

Citation addendum and correction to “The 2-Adic Sturmian Carry Constant in the Collatz Carry Equation”

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Elias De Jesús

Independent Researcher

ORCID: 0009-0007-0190-9143

dejesuseliass10@gmail.com

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Purpose

This addendum accompanies the note *The 2-Adic Sturmian Carry Constant in the Collatz Carry Equation* [2] (hereafter **the note**) without altering it. It supplies a citation the note omitted, qualifies the one priority claim that omission made unsustainable, and corrects a typographical slip in a proof together with a range hypothesis the proof uses but the theorem statement did not record.

No theorem, proof or computed value in the note changes. The single mathematical edit is the misprinted factor identified in Section 5, which has no consequence for any statement in the note. This addendum makes no new mathematical claim.

1 The omitted citation, and what it contains

J. López and P. Stoll, *The $3x+1$ conjugacy map over a Sturmian word*, *Integers* **9** (2009), #A13, 141–162 [3].

For every irrational slope $\alpha \in (0, 1)$ with convergents p_k/q_k , and with $1c_\alpha(j) = \lceil (j+1)\alpha \rceil - \lceil j\alpha \rceil$, their Theorem 1 gives, in $\mathbb{Z}_2$,

$$\Phi(1c_\alpha) = -\frac{1}{3} - \sum_{j \geq 0} (-1)^{j+1} \frac{2^{q_{j+1}+q_j-1}}{3(3^{p_{j+1}} - 2^{q_{j+1}})(3^{p_j} - 2^{q_j})}, \quad (1)$$

where Φ is the Bernstein–Lagarias conjugacy map [1]. Their Corollary 2 gives a generalized continued fraction for $-1/\Phi(1c_\alpha)$, convergent in $\mathbb{Z}_2$. Their §4 leaves $\mathbb{Z}_2$ and studies the real function $\Phi_{\mathbb{R}}$, proving irrationality of $\Phi_{\mathbb{R}}(m_\alpha)$ on $\mathbb{Q} \cap (\ln 2/\ln 3, 1]$ and of a dual limit point below $\ln 2/\ln 3$; the restriction is forced, since $\sum 2^{nq}/3^{np}$ converges in $\mathbb{R}$ if and only if $\ln 2/\ln 3 < p/q \leq 1$.

The denominators of (1) are the walls of two consecutive convergent shells, and the term exponents are $q_{j+1} + q_j - 1$. **These are the approximants and the approximation depths of the note.**

The companion paper [4] proves aperiodicity for parity words of ones-density strictly above $\ln 2/\ln 3$, and its Theorem 1 shows that a rational divergent trajectory would have density exactly $\ln 2/\ln 3$.

2 Dictionary

Signs were checked term by term against the note’s own Definition 4.1, and the identity was verified numerically modulo 2^{3000} .

	the note	López–Stoll
slope	$\alpha = \log_2 3 \approx 1.58496$	$\alpha_{\text{LS}} = \ln 2 / \ln 3 = \log_3 2 \approx 0.630930$
relation	$\alpha_{\text{LS}} = 1/\alpha$	
word	c_α , with $s_j(c_\alpha) = \lfloor j\alpha \rfloor$	$1c_{\alpha_{\text{LS}}}(j) = \lceil (j+1)\alpha_{\text{LS}} \rceil - \lceil j\alpha_{\text{LS}} \rceil$
constant	$\Xi_\alpha = \sum_{j \geq 0} 3^{-(j+1)} 2^{\lfloor j\alpha \rfloor}$	$\Phi(1c_{\alpha_{\text{LS}}})$
identity	$\Xi_\alpha = -\Phi(1c_{\ln 2 / \ln 3})$	
shell	$(q_n, p_n) = (\text{length}, \text{discharge})$	$(p_k, q_k) = (\text{numerator}, \text{denominator})$
relation	$(q_n, p_n) = (p_k, q_k)$: their numerator is the note’s word length	
approximant	$x_n = Q_n = -C_n/\delta_n$	$F(p_k/q_k) = \phi(m_{p/q})/(2^q - 3^p)$

Sign anchor. §9 of the note records the shell $(2, 3)$ as $Q = -5$ with $\Xi((1, 2)^\infty) = C/\delta = 5/1 = 5$; the convention above reproduces this. *A qualification on the approximants.* Their Definition 26 sets $F(p/q) = \phi(m_{p/q})/(2^q - 3^p)$, which is the periodic cycle value $\Phi(m_{p/q}^\infty)$. But the proof of their Lemma 27 gives $-P_{2k+1}/Q_{2k+1} = F(p_{2k+1}/q_{2k+1})$

while $-P_{2k}/Q_{2k} = F(p_{2k}/q_{2k}) + g(p_{2k}/q_{2k})$. **Only the odd-indexed continued-fraction convergents are periodic cycle values**; the even ones carry an additional gap term.

Depth correspondence. In the note’s coordinates the term exponents $q_{j+1} + q_j - 1$ of (1) read $p_n + p_{n+1} - 1$. Direct computation confirms this: the partial sums of (1) have error valuations 2, 4, 10, 26, 83, 148, 568, 1538, $\dots$, i.e. $p_n + p_{n+1} - 1$ for $n = 1, 2, 3, \dots$ in the numbering of §§9–10 of the note. The note’s lower-convergent values 10, 83, 568 are exactly the odd-indexed members of that sequence. Its upper-convergent values $p_n - 1$ (7, 64, 484) do *not* appear there: (1) records the valuation of its own tail, not $v_2(x_n + \Xi_\alpha)$ shell by shell.

3 Qualified claim

Original wording (§1, second paragraph).

“This note begins the study of the tower itself, and presents its first theorem-grade structural result.”

Corrected wording.

“This note begins the study of the tower itself, and presents a theorem-grade structural result for it. The result is not the first 2-adic theorem about this object. For every irrational slope, López and Stoll [3] give a closed 2-adic series for the Bernstein–Lagarias conjugacy value [1] of the characteristic Sturmian word, indexed by consecutive pairs of continued-fraction convergents; in the coordinates of this note its term exponents are $p_n + p_{n+1} - 1$, the lower-convergent depths of Theorem 7.1. What follows is an independent derivation, sharper in one respect: it separates the upper and lower convergent cases and pins each constant exactly, shell by shell.”

The abstract carries the same qualification. The proof given in the note is independent of [3] and was re-derived in full. Correspondingly, Remark 5.2, which credits [5] for the classical word-agreement fact, should add that the 2-adic consequence of that agreement — the periodic values of the convergent blocks approximating the conjugacy value of the characteristic word, with depths governed by consecutive convergent pairs — is Theorem 1 of [3].

4 Range of Theorem 7.1

Theorem 7.1 is stated in the note without a range. Its proof, through Lemma 5.1, uses $q_n \geq 2$ (the index range $1 \leq i \leq q_n - 1$ must be non-empty) and $q_{n+1} > q_n$. In the numbering of §§9–10, where $(q_1, p_1) = (1, 1)$, both hold from the shell (2, 3) onward, so **Theorem 7.1 holds as proved for $n \geq 3$** . The two initial shells (1, 1) and (1, 2) have $q_n = 1$ and lie outside the proof’s scope. The stated formula nevertheless returns the correct valuations there, 2 and 1 respectively, as verified by direct computation; this is an observation, not an extension of the theorem. Nothing in the note uses the two excluded shells.

5 Correction

Original (§5, proof of Lemma 5.1, lower case).

“At $j^* = q_n + q_{n+1}$ the mismatch occurs: $m_{j^*} = m_{q_{n+1}} = q_n - 1$ and $m_{j^*} + j^*|R_n| = q_n + q_n|R_n| - |R_{n+1}| > q_n$.”

Corrected.

“At $j^* = q_n + q_{n+1}$ the mismatch occurs: $m_{j^*} = m_{q_{n+1}} = q_n - 1$ and $m_{j^*} + j^*|R_n| = q_n + q_n|R_n| - q_n|R_{n+1}| > q_n$.”

Indeed, with $m_{j^*} = q_n - 1$ and $j^* = q_n + q_{n+1}$,

$$m_{j^*} + j^*|R_n| = (q_n - 1) + q_n|R_n| + q_{n+1}|R_n| = (q_n - 1) + q_n|R_n| + (1 - q_n|R_{n+1}|) = q_n + q_n|R_n| - q_n|R_{n+1}|,$$

using $q_{n+1}|R_n| + q_n|R_{n+1}| = 1$, equation (1) of the note. The factor q_n on the last term was dropped in the printed line.

Consequence: none. The inequality $> q_n$ follows from $|R_n| > |R_{n+1}|$, which is classical. Lemma 5.1, Theorem 7.1, Proposition 8.1 and every value in the computational ledger of §10 are unaffected.

Acknowledgment. The omission was identified during an adversarial citation audit of the Sturmian thread, assisted by Claude (Anthropic). All statements here, and any errors, are the author’s.

References

- [1] Daniel J. Bernstein and Jeffrey C. Lagarias. The $3x+1$ Conjugacy Map. *Canadian Journal of Mathematics*, 48(6):1154–1169, 1996.
- [2] Elías De Jesús. The 2-Adic Sturmian Carry Constant in the Collatz Carry Equation, 2026. technical note, 5 June 2026.
- [3] Josefina López and Peter Stoll. The $3x+1$ Conjugacy Map over a Sturmian Word. *Integers*, 9(A13):141–162, 2009.
- [4] Josefina López and Peter Stoll. The $3x+1$ Periodicity Conjecture in $\mathbb{R}$, 2021.
- [5] M. Lothaire. *Algebraic Combinatorics on Words*, volume 90 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2002. Chapter 2: Sturmian words.